



SubnettX

Zero-Trust Dual-Vector Attendance & Threat-Monitoring

INITIALIZING SYSTEM PROTOCOLS ...

PRESENTER SCRIPT

"Welcome, everyone. Today, we are introducing SubnettX. Our mission was simple but critical: to solve the fundamental flaw in modern attendance tracking—the inability to cryptographically and physically prove a student is in a specific room at an exact moment."

THE FLAWS IN LEGACY SYSTEMS



Manual Roll Call

Wastes 10–15 minutes of valuable lecture time and is highly prone to human error and proxy attendance.



Static QR Codes

Easily bypassed. Students photograph the QR, send it via WhatsApp, and friends scan it from their dorm rooms.



GPS & Geofencing

Vulnerable to basic spoofing techniques using fake GPS applications or VPN tunneling.



Bluetooth Beacons

Requires expensive, proprietary hardware installations in every classroom and drains student phone batteries.

PRESENTER SCRIPT

"We looked at traditional systems and realized they fail on multiple fronts. Digital verification isn't enough, and physical location is easily spoofed. We needed a paradigm shift."

ZERO-TRUST DUAL-VECTOR ARCHITECTURE

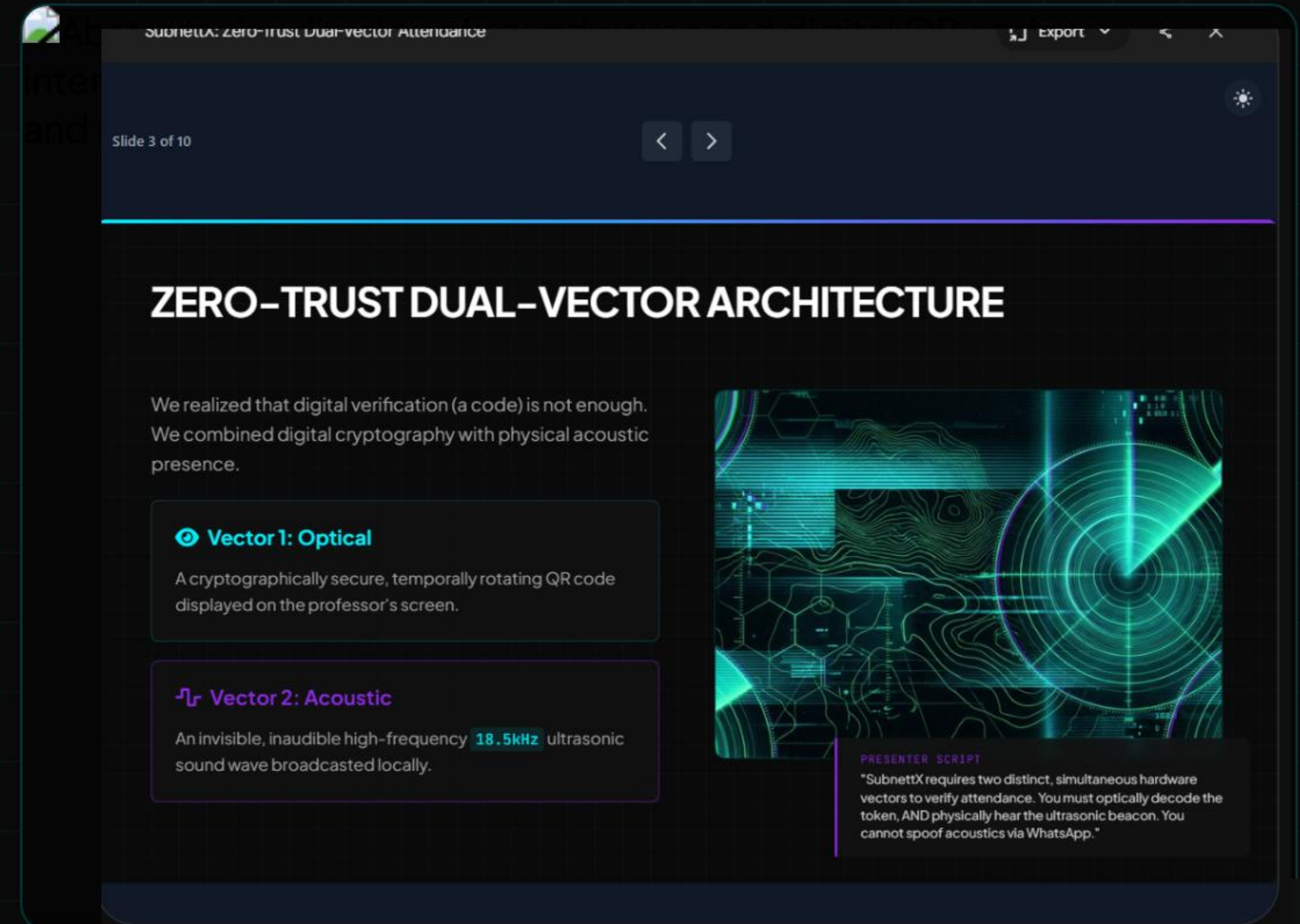
We realized that digital verification (a code) is not enough. We combined digital cryptography with physical acoustic presence.

👁 Vector 1: Optical

A cryptographically secure, temporally rotating QR code displayed on the professor's screen.

🔊 Vector 2: Acoustic

An invisible, inaudible high-frequency **18.5kHz** ultrasonic sound wave broadcasted locally.



"SubnettX requires two distinct, simultaneous hardware vectors to verify attendance. You must optically decode the token, AND physically hear the ultrasonic beacon. You cannot spoof acoustics via WhatsApp."

THE 4-SECOND VERIFICATION PIPELINE

STEP_01

The Broadcast

Professor opens the Dashboard. Laptop broadcasts a silent **18.5kHz** ultrasonic beacon & displays a rotating QR code.

STEP_02

Optical Scan

Student points phone camera at the QR code using the SubnettX Portal.

STEP_03

Mic Trigger

Moment the camera decodes the QR token, the phone triggers the mic via HTML5 Web Audio API.

STEP_04

FFT Scan

Fast Fourier Transform (FFT) algorithm scans the audio spectrum for the **18.5kHz** frequency bin.

STEP_05

Acoustic Handshake

If amplitude spikes above threshold, the acoustic handshake is locally successful.

STEP_06

Payload Delivery

System bundles Optical + Acoustic flags to the Flask backend within the strict temporal window.

DEEP DIVE: FAST FOURIER TRANSFORM

18.5

KILOHERTZ (KHZ)

INAUDIBLE TO HUMAN EAR

Signal Processing on the Edge

The student's mobile browser runs real-time signal processing. It isolates the specific frequency bin to detect the professor's laptop beacon, completely ignoring standard classroom noise.

Amplitude 18.5kHz > 25 ⇒ VERIFIED

*Threshold dynamically adjusted based on hardware calibration.

THREAT MITIGATION & BENEFITS

Absolute Proof of Presence



If a student scans a screenshot sent by a friend, their phone's microphone will NOT hear the ultrasonic beacon in the classroom. Handshake fails instantly.

Zero Proprietary Hardware (\$0 Deployment)



Utilizes standard laptop speakers and student phone microphones. No BLE beacons or RFID scanners required for the university to purchase.

Instantaneous Speed (SQLite WAL)



Entire handshake takes < 4 seconds. Our backend uses Write-Ahead Logging (WAL) concurrency, allowing 50 students to hit the server simultaneously without locking.

THE THREAT DASHBOARD

We didn't just build an attendance tracker; we built a live security dashboard using long-polling updates.



Roll Number: CS-1042

Optical: PASS | Acoustic: PASS



Roll Number: CS-1088 (Spoof Attempt)

Optical: PASS | Acoustic: FAILED

WELCOME TO THE THREAT DASHBOARD // SEED: A9EB5FEE

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SCAN ARRAY

SCANNING

THREAT LEVEL: LOW

OBJECTS IN RANGE: 004

REPORT

SYNC

ALERT

SECURITY SYNC

FIREWALL INTEGRITY

85%

WELCOME TO THE THREAT DASHBOARD

SYSTEM EVOLUTION: FUTURE SCOPE

Slide 8 of 10

SYSTEM EVOLUTION: FUTURE SCOPE



Machine Learning

Analyzing check-in time deltas and acoustic anomalies to find predictive patterns in student tardiness and proxy attempts.



Wasm Edge-Compute

Moving FFT acoustic processing entirely to WebAssembly (Wasm) for near-native speed and drastically lower mobile battery consumption.



Biometric Anchoring

Tying the cryptographic token to the device's secure hardware enclave (FaceID) to prevent credential sharing.

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System Secured.

The future of institutional verification is Dual-Vector.

We are now open for questions.

IMAGE SOURCES



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